

E-COMMERCE CUSTOMER BEHAVIOR ANALYSIS REPORT

*A Data-Driven Study of Purchasing Patterns, Customer Satisfaction,
and Retention Across an Online Retail Platform*

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Data Analytics Project | Business Analysis Report

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1. Executive Summary

This report analyzes 10,000 e-commerce transactions to uncover customer behavior across demographics, purchasing habits, payment methods, delivery performance, and satisfaction. It identifies a key retention challenge, with only one in five customers both satisfied and retained. Findings show delivery reliability has a greater impact than discounting, leading to recommendations on logistics, Premium subscriptions, targeted offers, and regional improvements.

2. Project Overview and Objectives

The purpose of this project is to analyze customer behavior on an online retail platform by identifying patterns that influence purchasing decisions, customer satisfaction, and retention. The analysis progresses through three levels: descriptive statistics to summarize customer behavior, relationship analysis to uncover key correlations, and advanced investigation to identify the primary factors driving customer churn and satisfaction.

2.1 Objectives

- Establish a clear statistical profile of the customer base, including age, spending, and engagement patterns.
- Identify which product categories, payment methods, and devices drive the greatest purchase volume.
- Quantify return-customer and satisfaction rates, and determine what factors are associated with each.
- Examine the relationship between website engagement (time spent), discounting, and purchase value.
- Assess regional variation in spending and delivery performance.
- Translate findings into specific, actionable business recommendations.

3. Dataset Description

The dataset comprises 10,000 individual customer records, each representing a single purchase transaction, with 16 attributes covering demographic, behavioral, transactional, and satisfaction-related information. The dataset contained no missing values and no duplicate rows, indicating a clean and reliable foundation for analysis.

3.1 Data Fields

Field	Description
Customer ID	Unique identifier for each customer record
Age	Customer age in years
Gender	Male, Female, or Other
Location	Customer's city (8 Bangladeshi divisions)
Product Category	One of 8 categories purchased
Purchase Amount (\$)	Transaction value in USD
Time Spent on Website (min)	Minutes spent browsing before purchase
Device Type	Mobile, Desktop, or Tablet
Payment Method	Method used to complete the transaction
Discount Availled	Whether a discount was applied (True/False)
Number of Items Purchased	Quantity of items in the transaction
Return Customer	Whether the customer has purchased before
Review Score (1-5)	Customer's rating of the transaction
Delivery Time (days)	Days taken to deliver the order
Subscription Status	Free, Trial, or Premium
Customer Satisfaction	Low, Medium, or High

4. Data Cleaning and Preparation

The dataset was inspected for quality and found to contain 10,000 complete records across 16 fields, with no missing values or duplicate transactions, requiring no data cleaning. To support quantitative analysis, the categorical Customer Satisfaction field was encoded into a numeric score (Low=1, Medium=2, High=3), enabling correlation and statistical analysis. No additional feature engineering was required.

5. Exploratory Data Analysis: Foundational Insights

This section summarizes the core descriptive statistics and distributional patterns found in the dataset, forming the foundation for the deeper analysis that follows.

5.1 Customer Age Profile

Metric	Value
Mean Age	43.79 years
Median Age	44 years
Mode (Most Frequent Age)	51 years

The age distribution is centered on middle-aged adults, indicating a financially stable customer base with strong purchasing power. The most common age (51) highlights a key segment that values reliability and classic products. Marketing should focus on dependable products and card-based promotional offers for this demographic.

5.2 Purchase Amount Variability

Metric	Value
Variance	81,924.05
Standard Deviation	\$286.22

A Z-score distribution of purchase amounts (visualized as a kernel density plot with outlier thresholds marked at plus/minus 2 standard deviations) shows that the large majority of transactions fall within a normal, consistent range. The distribution is relatively flat-topped rather than sharply peaked, meaning there is no single dominant purchase amount; spending is broadly and evenly distributed across the customer base. Density drops off sharply beyond the 2-standard-deviation mark, confirming that extremely low or high-value transactions are rare.

Business implication: Because spending behavior is consistent and evenly spread rather than concentrated around a few price points, broad-based discounting strategies are unlikely to be highly effective; targeted offers aligned to individual spending tiers would likely perform better.

5.3 Product Category Performance

Rank	Category	Purchase Count
1	Electronics	1,309
2	Clothing	1,297
3	Toys	1,288
4	Books	1,270
5	Home	1,265
6	Beauty	1,229
7	Groceries	1,193
8	Sports	1,149

Electronics slightly leads purchases, with Clothing and Toys close behind; all three are separated by only 21 transactions, showing broad customer interest rather than a dominant category. Beauty, Groceries, and Sports form a lower-performing tier, trailing by 80–160 transactions.

Recommendations

- **Inventory:** Prioritize stock management and alerts for Electronics, Clothing, and Toys.
- **Marketing:** Run seasonal promotions and style guides around these top categories.
- **Cross-selling:** Bundle top categories with lower-performing ones (e.g., Electronics + Beauty accessories) to increase visibility and sales.

5.4 Return Customers and Review Scores

Metric	Value
Return Customers	4,996 of 10,000 (approx. 50.0%)
Average Review Score	3.0 out of 5

Roughly half of all customers fall into the return-customer category, a relatively high proportion that points to friction somewhere in the post-purchase experience rather than an isolated issue. At the same time, the average review score of 3.0 is squarely neutral: customers are, on average, neither notably satisfied nor notably dissatisfied. Taken

together, these two figures suggest that product quality or delivery experience issues, rather than acute failures, are quietly eroding customer loyalty.

Recommendations

- Audit product condition both before dispatch and upon delivery to identify where quality issues originate.
- Launch a structured feedback or exit-survey campaign specifically targeting customers who have returned items, to diagnose root causes.

5.5 Subscription Status and Delivery Time

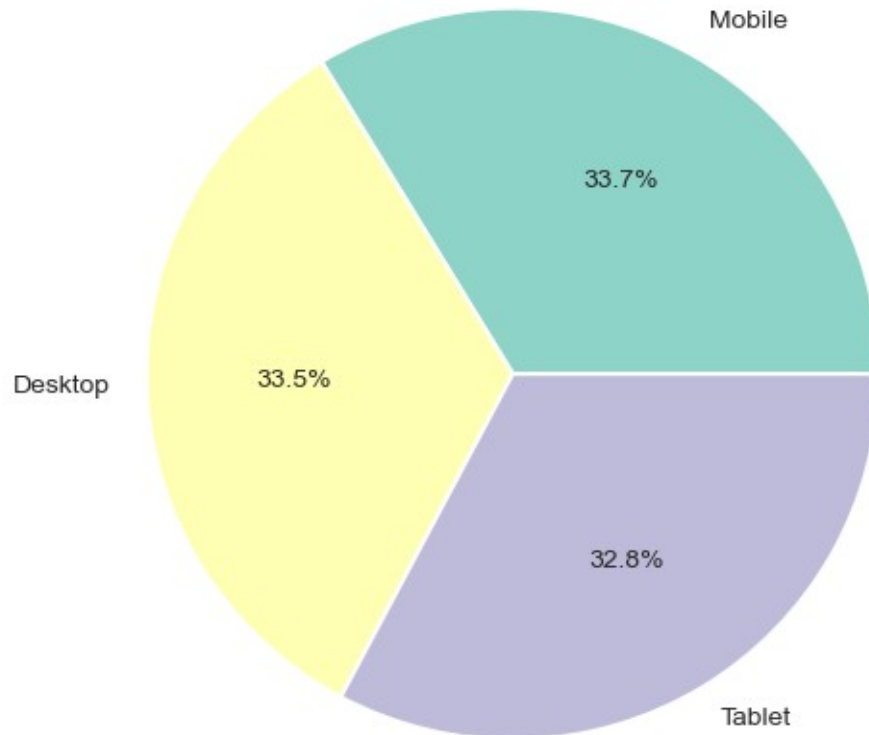
Metric	Value
Subscribed (Premium) Members	3,369 (33.7%)
Non-Subscribed Members	6,630 (66.3%)

Nearly two-thirds of customers are not subscribed to a paid membership tier. Critically, average delivery times for Free and Premium customers are similar (approximately 7 days for both), meaning non-subscribers receive comparable delivery service without paying for it. This undermines the incentive to upgrade and creates a risk that existing Premium subscribers may feel the paid tier delivers insufficient additional value, increasing the risk of subscription churn.

Recommendations

- Meaningfully differentiate delivery speed for Premium subscribers (e.g., guaranteed 1-2 day delivery) to justify the subscription cost and encourage upgrades from the large Free-tier base.
- Offer time-limited conversion incentives, such as 50% off the first month of Premium, paired with the improved delivery guarantee.

5.6 Device Usage Patterns

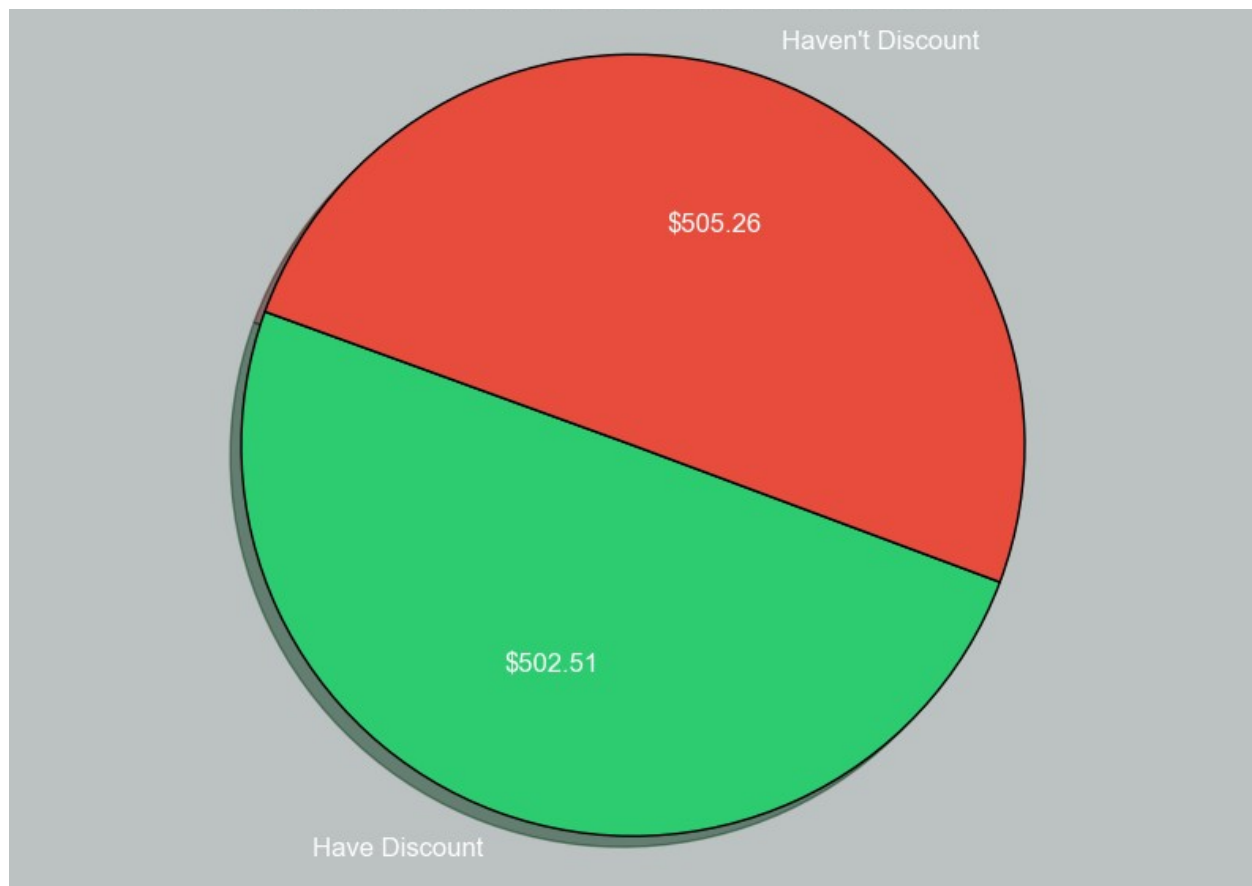


Purchase volume is evenly distributed across Mobile, Desktop, and Tablet devices, with Mobile accounting for **33.7%** and Tablet **32.4%**, showing consistent engagement across all platforms. Desktop users may favor bulk or work-time purchases, while Tablet usage suggests family or shared-device shopping.

Recommendations

- **UI/UX:** Maintain a seamless experience across all devices.
- **Mobile:** Expand one-tap payment options to reduce checkout friction.
- **Desktop:** Offer bulk-purchase incentives.
- **Tablet:** Promote bundle deals during seasonal shopping periods.

5.7 Discount-Driven Behavior

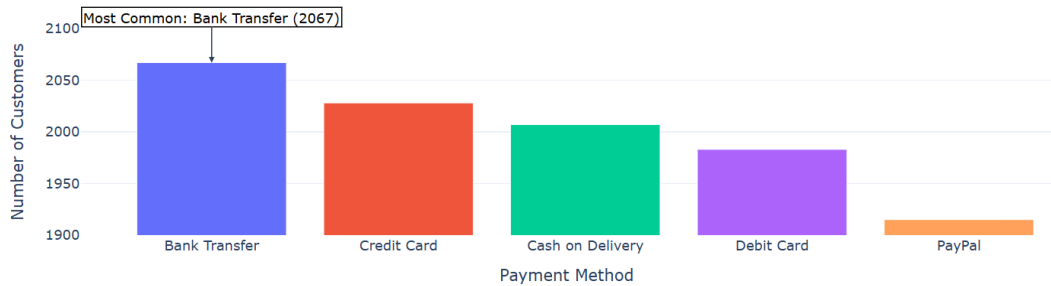


Close to half of all customers make purchases specifically when a discount is available, a pattern that is especially pronounced among Free-tier (non-subscribed) customers. This suggests a large segment of price-sensitive or new customers who are not yet engaging with the platform as a routine shopping destination, but rather opportunistically, in response to promotions.

Recommendations

- Move away from flat, single-product discounts, which do not meaningfully increase basket size, toward tiered discounts and combo offers that reward larger purchases.
- Introduce complementary product suggestions at checkout for discount-driven newcomers (e.g., suggesting a ring light to a customer purchasing a phone charger) to increase average order value and long-term engagement.

5.8 Payment Method Preferences



Bank Transfer is the most-used payment method, suggesting customers place a relatively high degree of trust in this channel, likely reflecting a tech-comfortable, middle-aged customer base consistent with the demographic profile in Section 5.1. PayPal is the least-used method, which may point to limited regional adoption or perceived complexity. Credit Card usage is also strong, indicating a segment of customers comfortable with deferred payment ("buy now, pay later") behavior, potentially salaried professionals.

Recommendations

- Improve the Bank Transfer user experience and pursue partnerships with the most-used banks to offer exclusive customer promotions.
- Increase engagement incentives for Credit Card users through targeted promotions.
- Explore delivery and marketing offers concentrated in areas with high workplace density, which may improve conversion and reduce returns for office-related orders.

6. Relationship and Correlation Analysis

This section moves beyond single-variable statistics to examine relationships between key behavioral metrics, engagement, satisfaction, and spending, to better understand what actually drives customer value.

6.1 Cash on Delivery and Customer Satisfaction

Payment Method	Avg. Review Score
Cash on Delivery	3.02

Payment Method	Avg. Review Score
PayPal	3.00
Debit Card	3.00
Bank Transfer	2.99
Credit Card	2.97

Although Cash on Delivery narrowly holds the highest average review score among payment methods, all five methods cluster tightly around the neutral midpoint of 3.0, indicating no single payment method delivers a meaningfully better customer experience. Given that Cash on Delivery is also the most commonly associated with first-time buyers, the fact that its score is not meaningfully higher than card- or transfer-based methods suggests unresolved friction, likely tied to delivery delays, product condition on arrival, or payment-collection hassles at the doorstep.

Recommendation

- Improve the Cash on Delivery experience by addressing communication gaps and delivery delays, particularly for first-time customers.
- Build trust through visible features such as easy-refund guarantees, verified order tracking, and displayed customer reviews at checkout.

6.2 Website Engagement vs. Purchase Amount

Metric	Value
Correlation Coefficient	0.01

The correlation between time spent on the website and purchase amount is effectively zero (0.01), indicating no meaningful relationship between how long a customer browses and how much they ultimately spend. This is a notable finding: customers appear to make purchase decisions quickly rather than through extended browsing and comparison, which may point to either efficient product discovery or, alternatively, a lack of engaging content that would otherwise encourage deeper exploration and larger baskets.

Recommendations

- Strengthen personalized product recommendations to convert browsing time into incremental purchases.
- Introduce smart search, filtering, and "customers also bought" features to encourage additional discovery and cross-selling during a session.

6.3 Satisfaction and Retention Overlap

Only approximately 20% of customers are both satisfied (rating of 4 or 5) and classified as return customers, meaning just one in five customers represents a genuinely loyal, happy repeat buyer. This is a critical retention signal: the majority of return customers are not necessarily satisfied ones, suggesting some repeat purchases may be driven by necessity, lack of alternatives, or promotional pull rather than genuine loyalty. Free-tier customers, who may expect Premium-level service without paying for it, are a plausible contributor to this gap.

Recommendation

- Conduct systematic quality review of products both before and after delivery to reduce the volume of dissatisfied repeat buyers.
- Launch a re-engagement campaign specifically targeting unsatisfied but returning customers, since this group represents meaningful upside if satisfaction can be improved.

6.4 Purchase Quantity and Satisfaction

There is no observable relationship between the number of items purchased in a transaction and the customer's reported satisfaction level. In other words, customers who buy more do not report being happier; satisfaction appears to be driven by factors other than basket size, such as delivery experience or product quality.

Recommendations

- Deploy a lightweight, platform-wide survey to systematically capture why customers are or are not satisfied, since basket size alone does not explain the pattern.
- Offer return incentives (such as loyalty discounts) to encourage dissatisfied customers to give the platform another opportunity.

6.5 Regional Performance: Barisal

Barisal ranks as the second-highest location by average purchase amount, trailing only the top-performing region, and outperforming five of the remaining six locations by a meaningful margin. Barisal customers also spend roughly \$11 more on average than customers who do not use discounts platform-wide, suggesting that Barisal's elevated spending is not primarily discount-driven. This points to other factors, such as service quality, brand trust, or a concentration of higher-income or premium customers, as more likely explanations.

Business implication: Barisal represents an attractive market for targeted marketing and loyalty investment, since its customers appear to spend based on value and experience rather than promotional pricing alone.

7. Critical Thinking: Advanced Diagnostic Insights

This section synthesizes multiple variables together to identify the underlying drivers of customer returns, satisfaction, and regional performance, moving from what is happening to why it may be happening.

7.1 Drivers of Return-Customer Status

Product Category Effects

- Toys, Electronics, and Groceries show a higher share of returning customers relative to non-returning customers than other categories.
- Electronics and Clothing are both high-volume and high-return categories; for Electronics, this is plausibly linked to product sensitivity and unmet expectations on arrival.
- Toys, despite comparatively lower order volume, show a disproportionately high return association, possibly reflecting quality concerns relative to customer expectations.

Satisfaction Score Effects

- Customers with Low or Medium satisfaction scores are more likely to be classified as return customers than those with High satisfaction.
- Only the High-satisfaction segment shows a majority of non-returning purchases, indicating that unmet expectations are a widespread, not isolated, issue.

Engagement and Device Effects

- Customers who spend more time browsing the website are, somewhat counter-intuitively, more likely to become return customers rather than fully satisfied one-time buyers.
- Desktop users make up roughly one-third of all customers and tend to browse longer, but this extended engagement does not reliably translate into loyalty.

Taken together, these patterns suggest a platform-wide gap in interactivity and personalization, and limited differentiation between the Free and Premium experience, both of which may be dampening long-term customer loyalty despite reasonable engagement levels.

7.2 Payment Method Impact on Satisfaction and Returns

With the exception of Credit Card and PayPal users, every other payment method shows higher average satisfaction among non-returned customers than among returned ones, reinforcing that returns are generally linked to product and delivery experience rather than the payment method itself. Notably, Cash on Delivery satisfaction scores remain low across both returned and non-returned customer groups, reinforcing the Section 6.1 finding that COD requires targeted operational investigation and improvement.

7.3 Regional Purchase Amount and Delivery Time

Location	Avg. Purchase Amount (\$)
Khulna	513.94
Barisal	513.67
Mymensingh	507.89
Chittagong	507.85
Dhaka	502.00
Rajshahi	495.54
Sylhet	494.98
Rangpur	494.37

Khulna, Barisal, and Sylhet stand out as relatively high-spending regions despite experiencing longer average delivery times (in the 10-13 day range in the delivery-time comparison), while Rangpur and Mymensingh show comparatively lower spending regardless of delivery speed. This indicates that delivery speed alone does not determine regional spending; customer demand in Khulna, Barisal, and Sylhet appears resilient even under slower logistics, representing a high-value opportunity if service levels were improved.

Recommendations

- Prioritize fast-delivery optimization specifically in high-spending, currently slower-delivery regions such as Khulna, Barisal, and Sylhet.
- Extend expedited delivery guarantees to Premium customers in these regions first, where the return on logistics investment is likely highest.

- Avoid universal, platform-wide investment in fast delivery, since faster delivery does not consistently correlate with higher spend in every region; a location-by-location logistics strategy will be more cost-effective.

8. Key Insights Summary

- **A financially stable, middle-aged core customer base.** Average and median ages cluster around 44, with a secondary concentration near 51, pointing to a customer base with meaningful purchasing power and a preference for dependable, classic products.
- **Broad, evenly distributed product interest.** No single category dominates; Electronics, Clothing, and Toys form a closely matched top tier, indicating diverse cross-category demand rather than a niche focus.
- **A significant, unresolved retention gap.** With a roughly 50% return-customer rate but only about 20% of customers both satisfied and retained, loyalty is being driven more by necessity or habit than genuine satisfaction for a large share of the base.
- **Delivery experience, not discounting, is the likely lever for loyalty.** Time on site and basket size show no meaningful correlation with either spending or satisfaction, while payment method and regional delivery performance show clearer links to satisfaction and returns.
- **Cash on Delivery is an operational weak point.** COD is both the most first-time-buyer-associated payment method and one associated with comparatively low, unresolved satisfaction, a high-priority area for operational investigation.
- **The Premium subscription lacks differentiation.** Comparable delivery times between Free and Premium tiers reduce the incentive to upgrade and put existing subscriptions at risk of churn.
- **Regional demand outpaces regional service levels.** Khulna, Barisal, and Sylhet sustain high spending despite slower delivery, representing a clear opportunity for targeted logistics investment.

9. Business Recommendations

9.1 Strengthen the Cash on Delivery Experience

- **Problem:** COD is the most common payment method but consistently has low customer satisfaction.

- **Suggested Action:** Improve delivery reliability, payment handling, and add trust features like order tracking and easy refunds.
- **Expected Business Impact:** Better first-time customer experience, higher repeat purchases, and fewer returns.

9.2 Differentiate the Premium Subscription Tier

- **Problem:** Premium and Free users receive similar delivery times, reducing subscription value.
- **Suggested Action:** Offer exclusive 1-2 day delivery and promotional incentives for Premium members.
- **Expected Business Impact:** Higher Premium conversions, improved retention, and increased recurring revenue.

9.3 Shift to Targeted Promotions

- **Problem:** Flat discounts attract buyers but do not significantly improve spending or satisfaction.
- **Suggested Action:** Replace blanket discounts with tiered offers, bundles, and personalized recommendations.
- **Expected Business Impact:** Higher average order value and more efficient promotional spending.

9.4 Prioritize Logistics in High-Value Regions

- **Problem:** High-spending regions experience longer delivery times.
- **Suggested Action:** Focus logistics improvements and faster delivery in Khulna, Barisal, and Sylhet.
- **Expected Business Impact:** Increased satisfaction, stronger retention, and better logistics efficiency.

9.5 Improve Product Quality Assurance

- **Problem:** High return rates, especially in quality-sensitive categories.
- **Suggested Action:** Strengthen quality checks and collect structured feedback from returning customers.
- **Expected Business Impact:** Fewer returns, better customer reviews, and stronger loyalty.

9.6 Enhance Cross-Device Personalization

- **Problem:** Similar purchasing behavior across devices but limited engagement-driven upselling.

- **Suggested Action:** Improve cross-device UX with personalized recommendations and device-specific promotions.
- **Expected Business Impact:** Higher conversion rates, increased order values, and better customer engagement.

10. Conclusion

The analysis shows the platform has a financially stable and diverse customer base with balanced purchasing across products, devices, and regions. The main challenge is customer retention, as only one in five customers is both satisfied and retained. Delivery reliability and payment experience—especially for Cash on Delivery—have a greater impact than discounting. Improving logistics, strengthening Premium benefits, and investing in high-value regions can significantly increase long-term customer loyalty and profitability.